

```

import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("train.csv")

df.head()
print(df.info())
print(df.isnull().sum())
print(df.describe())
survival_count = df["Survived"].value_counts()

plt.figure(figsize=(6,4))

plt.bar(["Not Survived", "Survived"], survival_count)

plt.title("Survival Count")
plt.ylabel("Number of Passengers")

plt.show()
female_survival = df[df["Sex"] == "female"]["Survived"].mean()
male_survival = df[df["Sex"] == "male"]["Survived"].mean()

plt.figure(figsize=(6,4))

plt.bar(["Female", "Male"], [female_survival, male_survival])

plt.title("Survival Rate by Gender")
plt.ylabel("Survival Rate")

plt.show()
plt.figure(figsize=(8,5))

plt.hist(df["Age"].dropna(), bins=20)

plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")

plt.show()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null   int64
1   Survived     891 non-null   int64
2   Pclass       891 non-null   int64
3   Name         891 non-null   object
4   Sex          891 non-null   object
5   Age         714 non-null   float64
6   SibSp        891 non-null   int64

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7   Parch      891 non-null   int64
8   Ticket     891 non-null   object
9   Fare       891 non-null   float64
10  Cabin      204 non-null   object
11  Embarked   889 non-null   object

```

```
dtypes: float64(2), int64(5), object(5)
```

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memory usage: 83.7+ KB
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```
None
```

```
PassengerId      0
```

```
Survived         0
```

```
Pclass          0
```

```
Name            0
```

```
Sex             0
```

```
Age            177
```

```
SibSp           0
```

```
Parch           0
```

```
Ticket          0
```

```
Fare            0
```

```
Cabin          687
```

```
Embarked        2
```

```
dtype: int64
```

	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200



